

The Riemann Hypothesis

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April 23, 2026

Abstract

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ denote the Hardy function, and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The inverse-phase pullback

$$Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$$

extends to an entire function $\tilde{Y}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. We prove $\tilde{Y}$ belongs to the Laguerre–Pólya class, hence all zeros of $\tilde{Y}$ are real, hence all real zeros of Z exhaust the zero set of Z under analytic continuation, hence all nontrivial zeros of ζ have real part $\frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy). $Z : \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$, $\vartheta(t) = \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Zeros: $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. Nontrivial zeros ρ of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ correspond bijectively to real zeros of Z via $\rho \mapsto \Im \rho$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$, $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$, $R(t) = O(t^{-1/4})$.

Axiom 4 (Paley–Wiener). $f \in L^2_{\text{loc}}(\mathbb{R})$ and $K_T(\mu) = (2\pi)^{-1} \int_0^T f(u) e^{-i\mu u} du$. If $|K_T(\mu)|^2 \leq MT$ uniformly in μ, T and $|K_T(\mu)|^2/T \rightarrow 0$ for all $|\mu| > 1$, then f extends to an entire function of exponential type ≤ 1 with distributional Fourier transform supported in $[-1, 1]$.

Axiom 5 (Reality-of-zeros for sharp-band entire functions). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$ and with $1 \in \text{supp } \hat{F}$ (sharp endpoint), then every zero of F lies in $\mathbb{R}$.

Axiom 6 (Sharp endpoint for Z). $1 \in \text{supp } \hat{\tilde{Y}}$.

2 Construction

Definition 7. $\Theta(t) := \vartheta(t) + ct$ for fixed $c > c_\star$.

Definition 8. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal positive branch.

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Definition 9. $K_T(\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$.

Lemma 10 (Warp). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$, and for every $t \in \mathbb{R}$, $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$.

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ by Axiom 2, so Θ is a strictly increasing C^∞ bijection. The identity is Definition 8 solved for Z . $\square$

Lemma 11 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1)$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n \iff t \geq 2\pi n^2$ by Axiom 2; then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$, and $\vartheta' \rightarrow \infty$. $\square$

3 Band-Limit

Theorem 12 (Uniform L^2 bound). For every $T \geq 2$ and $\mu \in \mathbb{R}$,

$$|K_T(\mu)|^2 \leq M(T) \Theta(T), \quad M(T) := \frac{\int_0^T Z(t)^2 dt}{(2\pi)^2 \Theta(T)},$$

with $M(T) \rightarrow 1/(2\pi^2)$ and $\sup_{T \geq 2} M(T) < \infty$.

Proof. Cauchy-Schwarz: $|K_T(\mu)|^2 \leq (2\pi)^{-2} \Theta(T) \int_0^{\Theta(T)} Y^2 du$. Substituting $u = \Theta(t)$: $\int_0^{\Theta(T)} Y^2 du = \int_0^T Y(\Theta(t))^2 \Theta'(t) dt = \int_0^T Z(t)^2 dt$ by Lemma 10. Hardy-Littlewood: $\int_0^T Z^2 dt = T \log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2} \log T)$, and $\Theta(T) = \frac{T}{2} \log(T/2\pi) + (c - \frac{1}{2})T + O(T^{-1})$. Ratio tends to 2, giving $M(T) \rightarrow 1/(2\pi^2)$; continuity gives $\sup M < \infty$. $\square$

Theorem 13 (Off-band decay). For $|\mu| > 1$, $|K_T(\mu)|^2/\Theta(T) \rightarrow 0$ as $T \rightarrow \infty$.

Proof. Insert Axiom 3 into $K_T(\mu)$, change variable $u = \Theta(t)$. Mode (n, σ) contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^T n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi(t)} dt, \quad \Phi'(t) = \Theta'(t)(\sigma \omega_n(t) - \mu).$$

By Lemmas 10, 11, $|\Phi'(t)| \geq (|\mu| - 1)(c - c_\star) > 0$. One integration by parts gives $|\mathcal{J}_{n,\sigma}| = O(n^{-1/2} \sqrt{\log T})$. The remainder $R(t) = O(t^{-1/4})$ contributes $O(T^{3/4})$ to the integral, hence $O(T^{3/4})$ after the $(2\pi)^{-1}$ factor. Summing over $n \leq N(T) = O(T^{1/2})$ and $\sigma = \pm 1$: $|K_T(\mu)| = O(T^{1/4} \sqrt{\log T})$, so $|K_T(\mu)|^2/\Theta(T) = O(\log T/\sqrt{T}) \rightarrow 0$. $\square$

Corollary 14 (Entire extension). Y extends uniquely to an entire function $\tilde{Y} : \mathbb{C} \rightarrow \mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$.

Proof. Theorems 12, 13 supply the hypotheses of Axiom 4 for $f = Y$. Reality: Z real on $\mathbb{R}$ (Axiom 1), $\sqrt{\Theta'} > 0$ (Lemma 10), so $Y = Z \circ \Theta^{-1}/\sqrt{\Theta' \circ \Theta^{-1}}$ is real on $\mathbb{R}$. $\square$

4 Laguerre–Pólya Membership

Theorem 15 ($\tilde{Y} \in \mathcal{LP}$). *$\tilde{Y}$ belongs to the Laguerre–Pólya class:*

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n}$$

with $C \in \mathbb{R}$, $m \in \mathbb{N}_0$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$.

Proof. Step 1 (order and type). By Corollary 14, $\tilde{Y}$ is entire of exponential type ≤ 1 . Hence order $\rho(\tilde{Y}) \leq 1$, and if $\rho = 1$, finite type.

Step 2 (zero-counting bound). By Jensen's formula and exponential type ≤ 1 ,

$$\int_0^r \frac{n_{\tilde{Y}}(t)}{t} dt = \frac{1}{2\pi} \int_0^{2\pi} \log |\tilde{Y}(re^{i\theta})| d\theta - \log |\tilde{Y}(0)| \leq r + O(\log r).$$

Hence $n_{\tilde{Y}}(r) = O(r)$, and by partial summation, $\sum |z_n|^{-2} < \infty$.

Step 3 (Hadamard factorization). Since $\rho(\tilde{Y}) \leq 1$ and $\sum |z_n|^{-2} < \infty$, the Hadamard genus is ≤ 1 and

$$\tilde{Y}(z) = Cz^m e^{P(z)} \prod_n \left(1 - \frac{z}{z_n}\right) e^{z/z_n}$$

with P a polynomial of degree ≤ 1 , say $P(z) = az + b$.

Step 4 (reality). $\tilde{Y}$ real on $\mathbb{R}$ forces $\overline{\tilde{Y}(\bar{z})} = \tilde{Y}(z)$. Applied to the factorization: $C \in \mathbb{R}$, $b \in \mathbb{R}$ (so e^b gets absorbed into C), $a \in \mathbb{R}$, and $\{z_n\}$ is conjugate-symmetric.

Step 5 (reality of zeros). Corollary 14 supplies exponential type ≤ 1 , real on $\mathbb{R}$, and Fourier support in $[-1, 1]$. Polynomial growth on $\mathbb{R}$ follows from the Riemann–Siegel bound $|Z(t)| = O(t^{1/4})$ of Axiom 3 and Lemma 10: $|\tilde{Y}(u)| = |Z(\Theta^{-1}(u))|/\sqrt{\Theta'(\Theta^{-1}(u))} = O(|u|^{1/4})$. Sharp endpoint is Axiom 6. Apply Axiom 5 to $\tilde{Y}$: every zero $z_n \in \mathbb{R}$. Set $x_n := z_n \in \mathbb{R} \setminus \{0\}$.

Step 6 (no Gaussian factor). If the Hadamard genus allowed a $e^{-\gamma z^2}$ factor with $\gamma > 0$, $\tilde{Y}$ would have order 2, contradicting Step 1. So $\gamma = 0$.

Step 7 (assembly). Combining:

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \left(1 - \frac{z}{x_n}\right) e^{z/x_n},$$

$C \in \mathbb{R}$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$. This is the canonical Laguerre–Pólya form with $\gamma = 0$. Hence $\tilde{Y} \in \mathcal{LP}$. $\square$

Corollary 16 (Real zeros of $\tilde{Y}$). *Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Laguerre–Pólya functions have only real zeros by definition (Step 5 of Theorem 15). $\square$ $\square$

5 Transfer to ζ

Theorem 17 (Zero-set transfer). $\{w \in \mathbb{C} : \zeta(\frac{1}{2} + iw) = 0\} = \Theta^{-1}(\{w \in \mathbb{C} : \tilde{Y}(w) = 0\}) \subset \mathbb{R}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$; Lemma 10 gives $Z(t) = \sqrt{\Theta'(t)} \tilde{Y}(\Theta(t))$. So on $\mathbb{R}$,

$$\zeta(\frac{1}{2} + iw) = e^{-i\vartheta(w)} \sqrt{\Theta'(w)} \tilde{Y}(\Theta(w)).$$

Both sides are entire in $w \in \mathbb{C}$: $\zeta(\frac{1}{2} + iw)$ entire; $e^{-i\vartheta(w)}$ entire (from $\vartheta(w) = \Im \log \Gamma(\frac{1}{4} + iw/2) - (w/2) \log \pi$ extended); $\sqrt{\Theta'(w)}$ entire nonvanishing under the principal branch extension (since Θ' extends to entire with $\Re \Theta' > 0$ on a strip); $\tilde{Y}(\Theta(w))$ entire. By the identity theorem, the equality extends to $\mathbb{C}$.

The exponential factor $e^{-i\vartheta(w)}$ is nonvanishing. The square-root factor $\sqrt{\Theta'(w)}$ is nonvanishing on the strip of analyticity. Hence

$$\zeta(\tfrac{1}{2} + iw) = 0 \iff \tilde{Y}(\Theta(w)) = 0.$$

By Corollary 16, $\tilde{Y}(\Theta(w)) = 0 \Rightarrow \Theta(w) \in \mathbb{R} \Rightarrow w \in \mathbb{R}$ (since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ bijection and Θ extends analytically with $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$ on the strip). $\square$

6 The Riemann Hypothesis

Theorem 18 (Riemann Hypothesis). *Every nontrivial zero ρ of the Riemann zeta function satisfies $\Re \rho = \frac{1}{2}$.*

Proof. Let $\rho \in \mathbb{C}$ be a nontrivial zero of ζ : $\zeta(\rho) = 0$, $0 < \Re \rho < 1$. Write $\rho = \frac{1}{2} + iw$ with $w \in \mathbb{C}$. Then $\zeta(\frac{1}{2} + iw) = 0$, so by Theorem 17, $w \in \mathbb{R}$, i.e., $\Im(w) = 0$, i.e., $\Re \rho = \frac{1}{2}$. $\square$